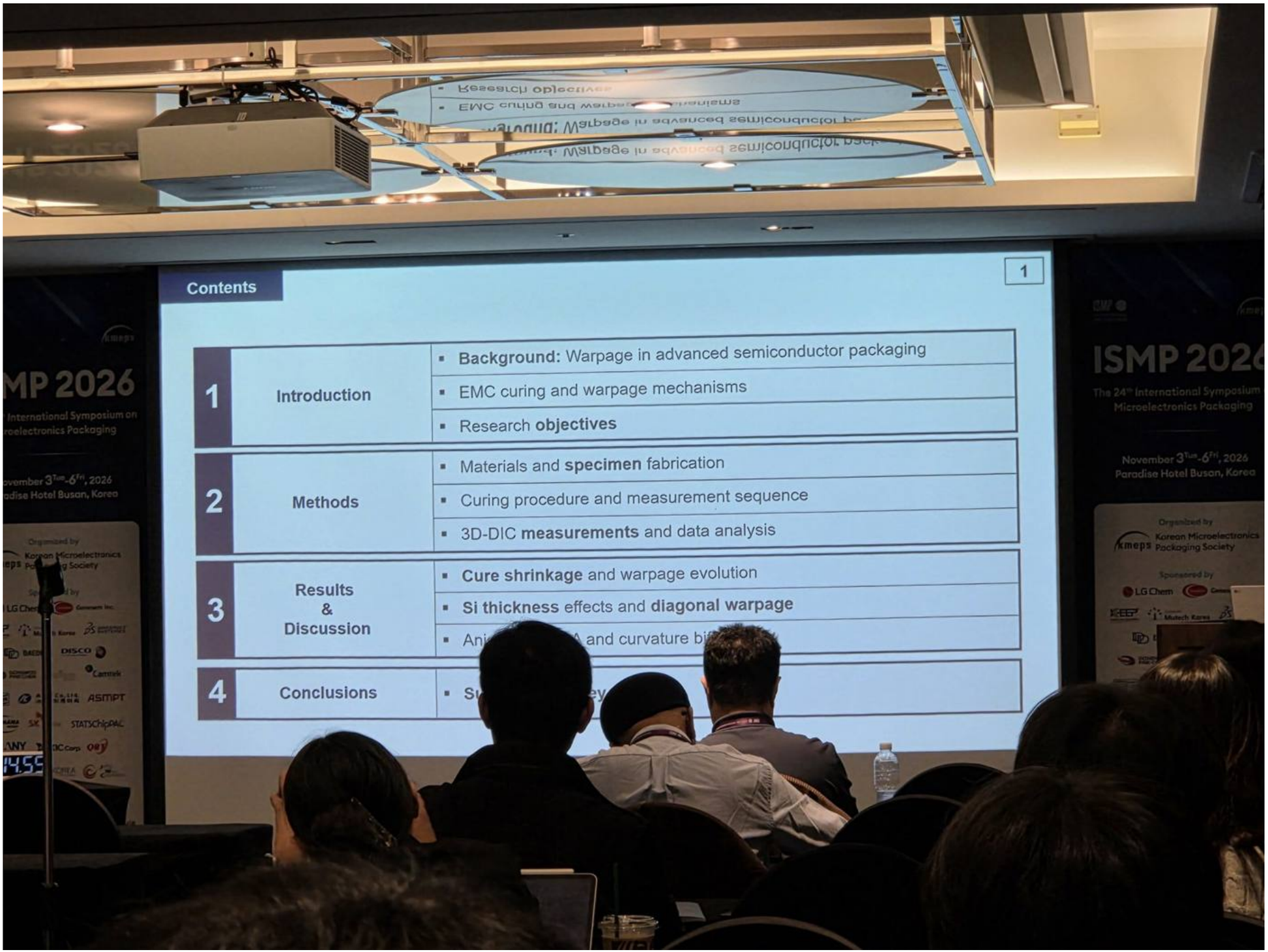
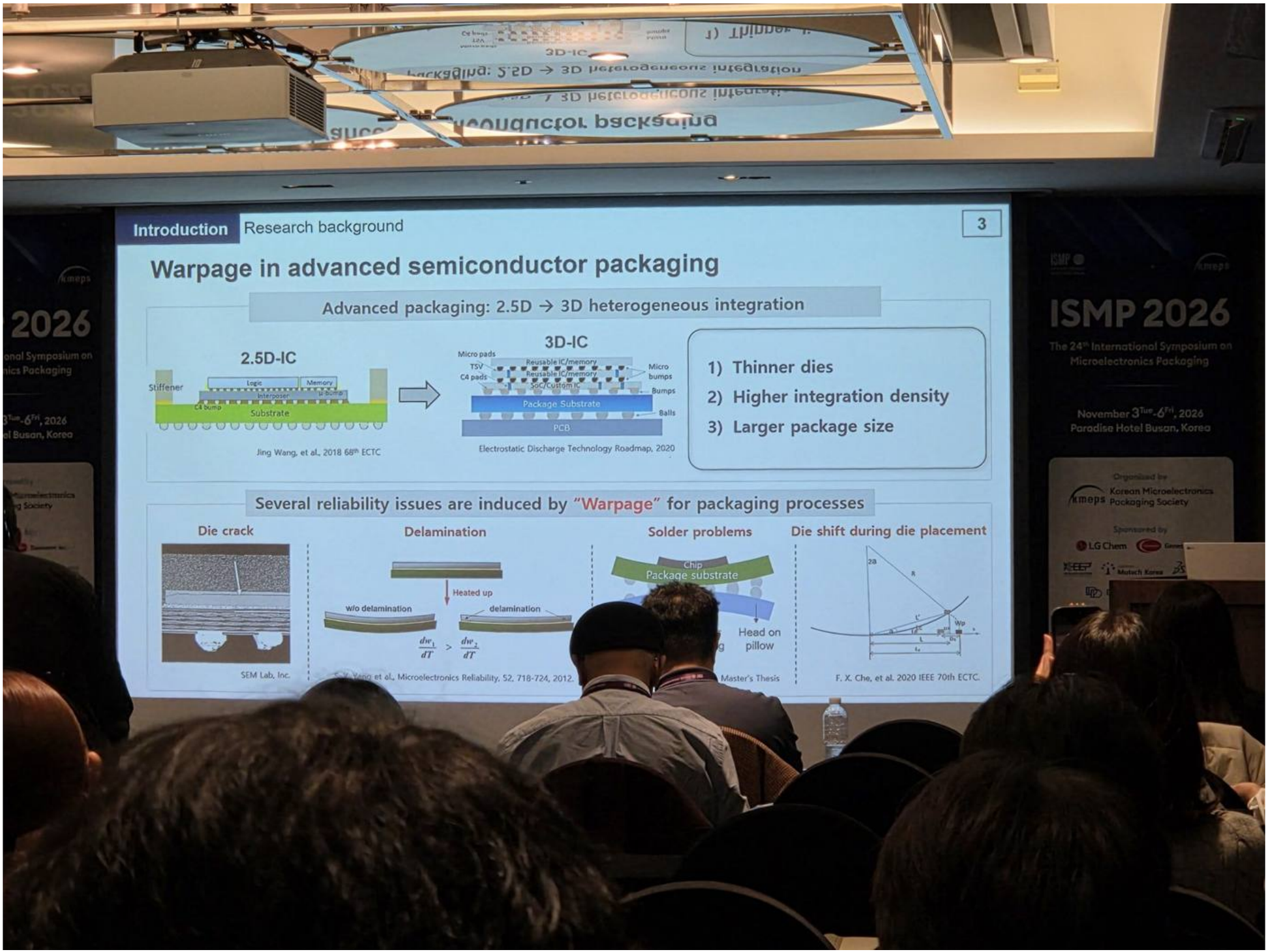
T13 EMC 경화수축과 얇은 실리콘의 휨 분기

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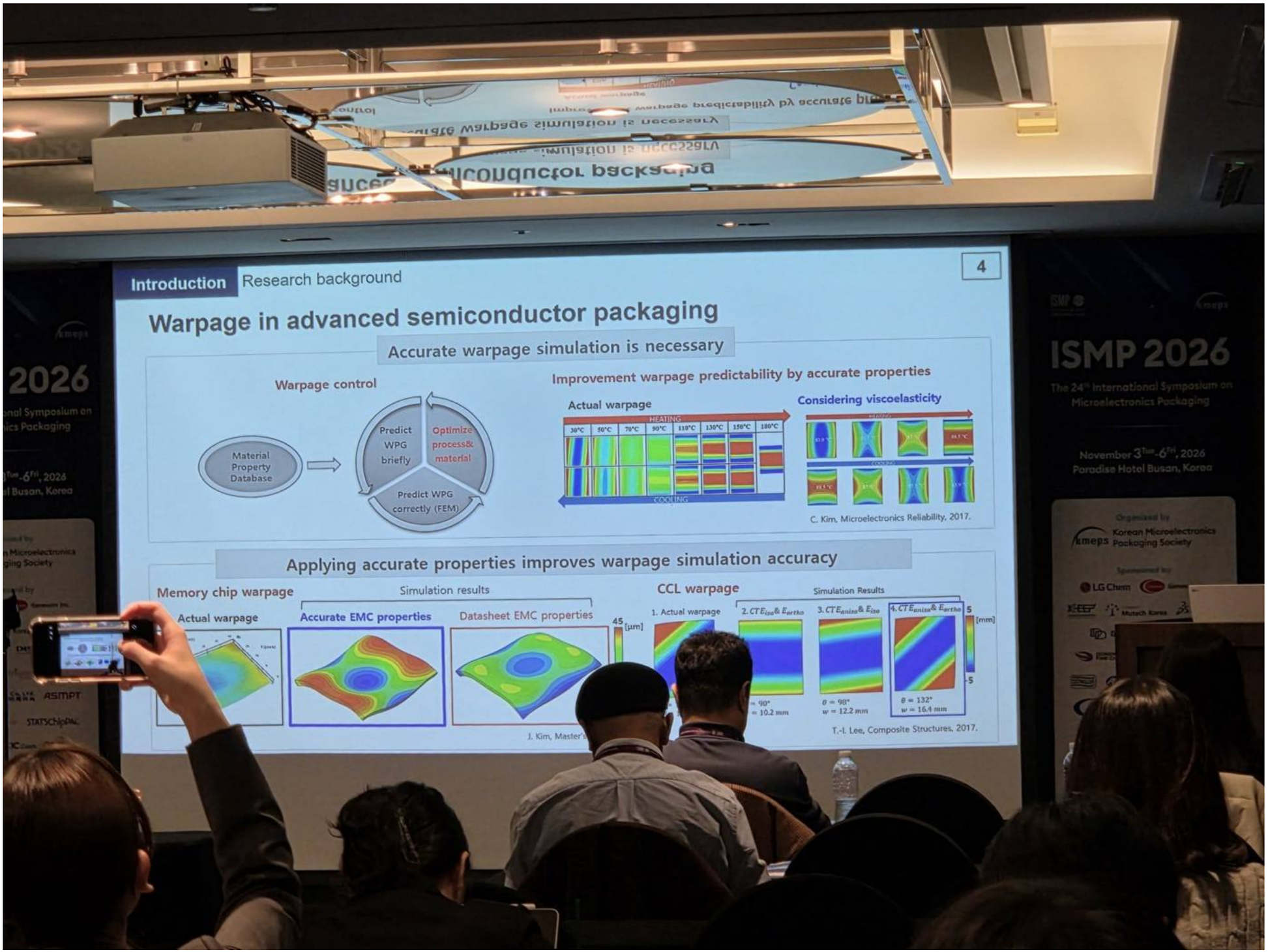
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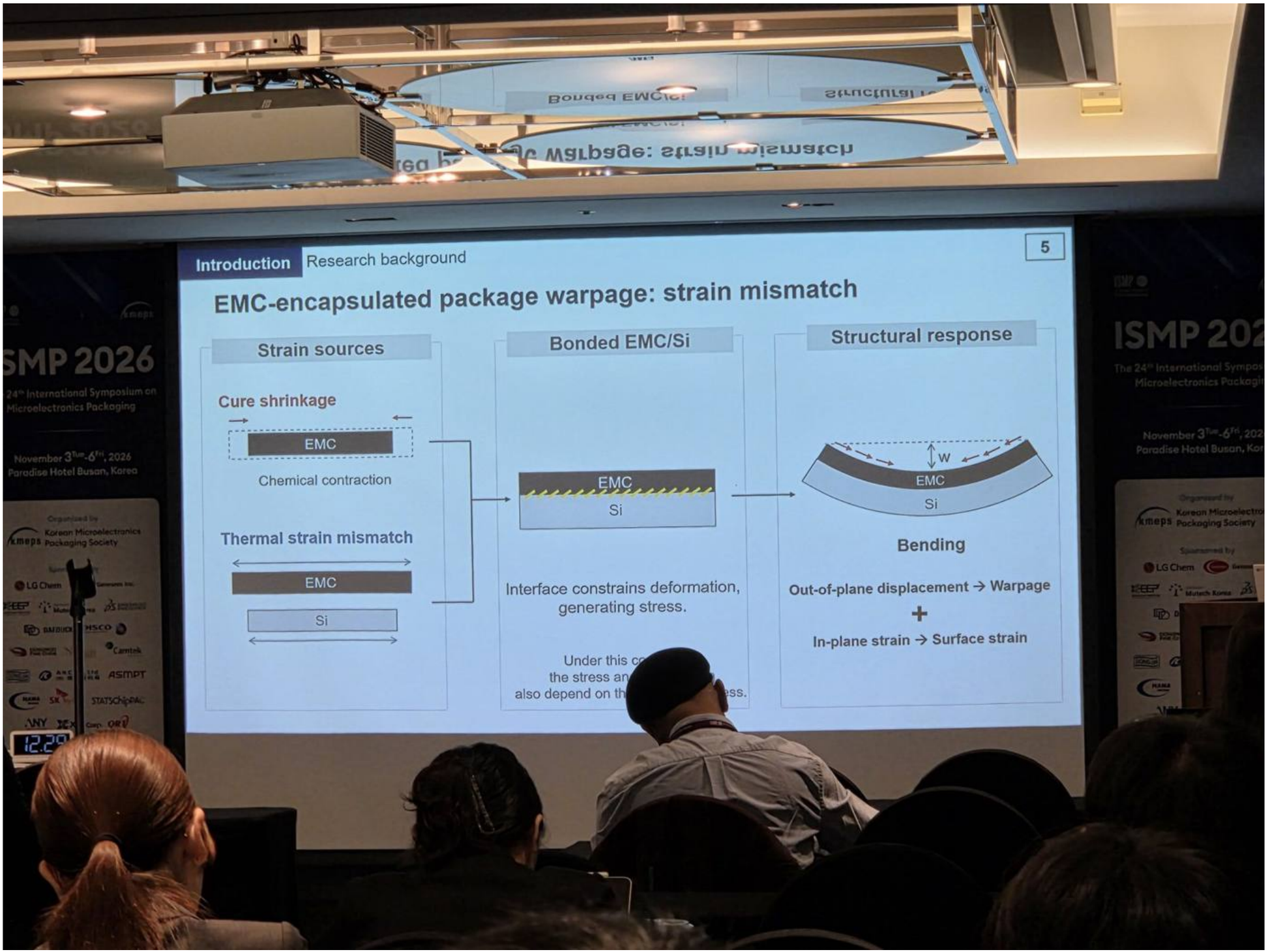
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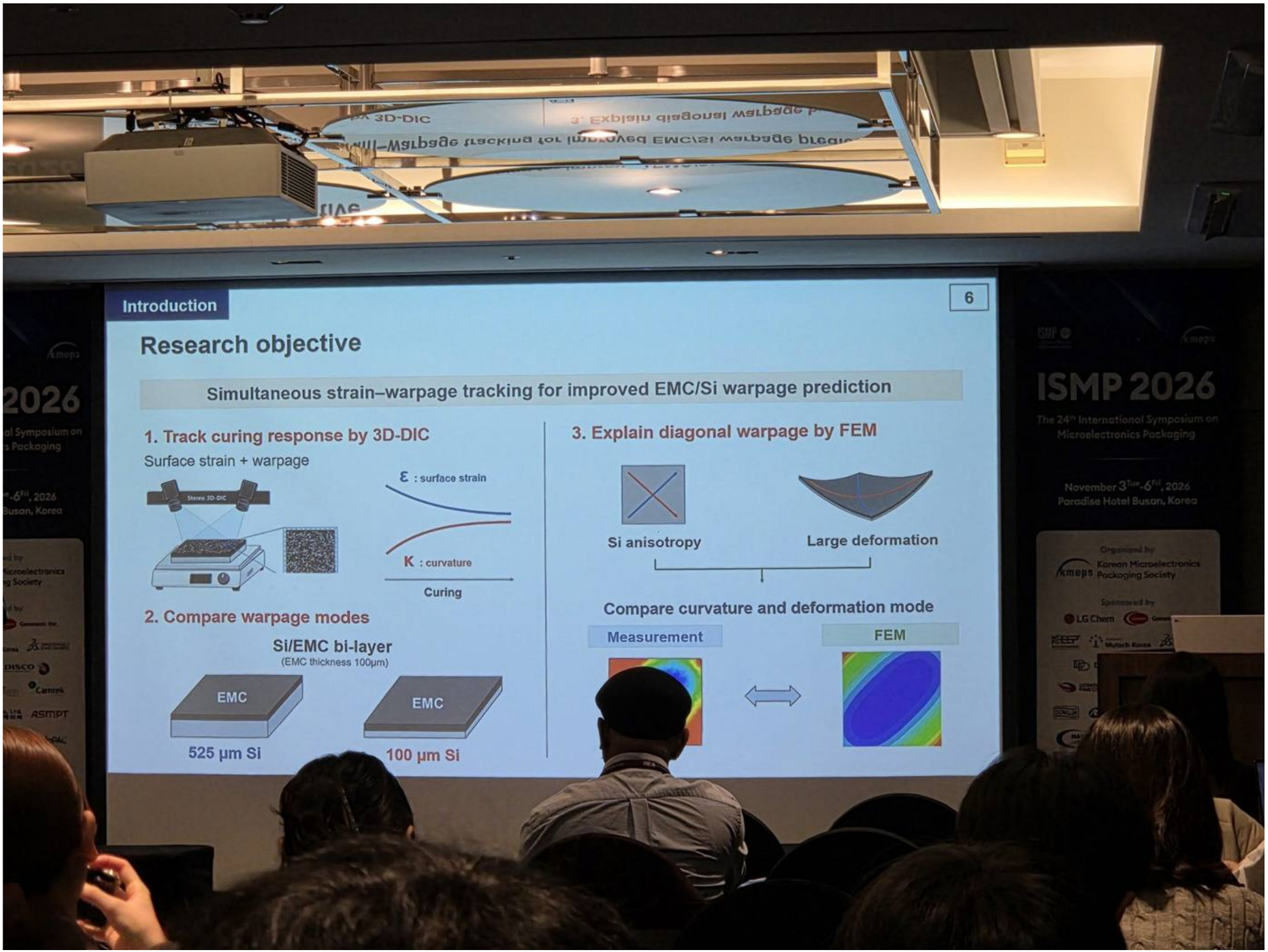
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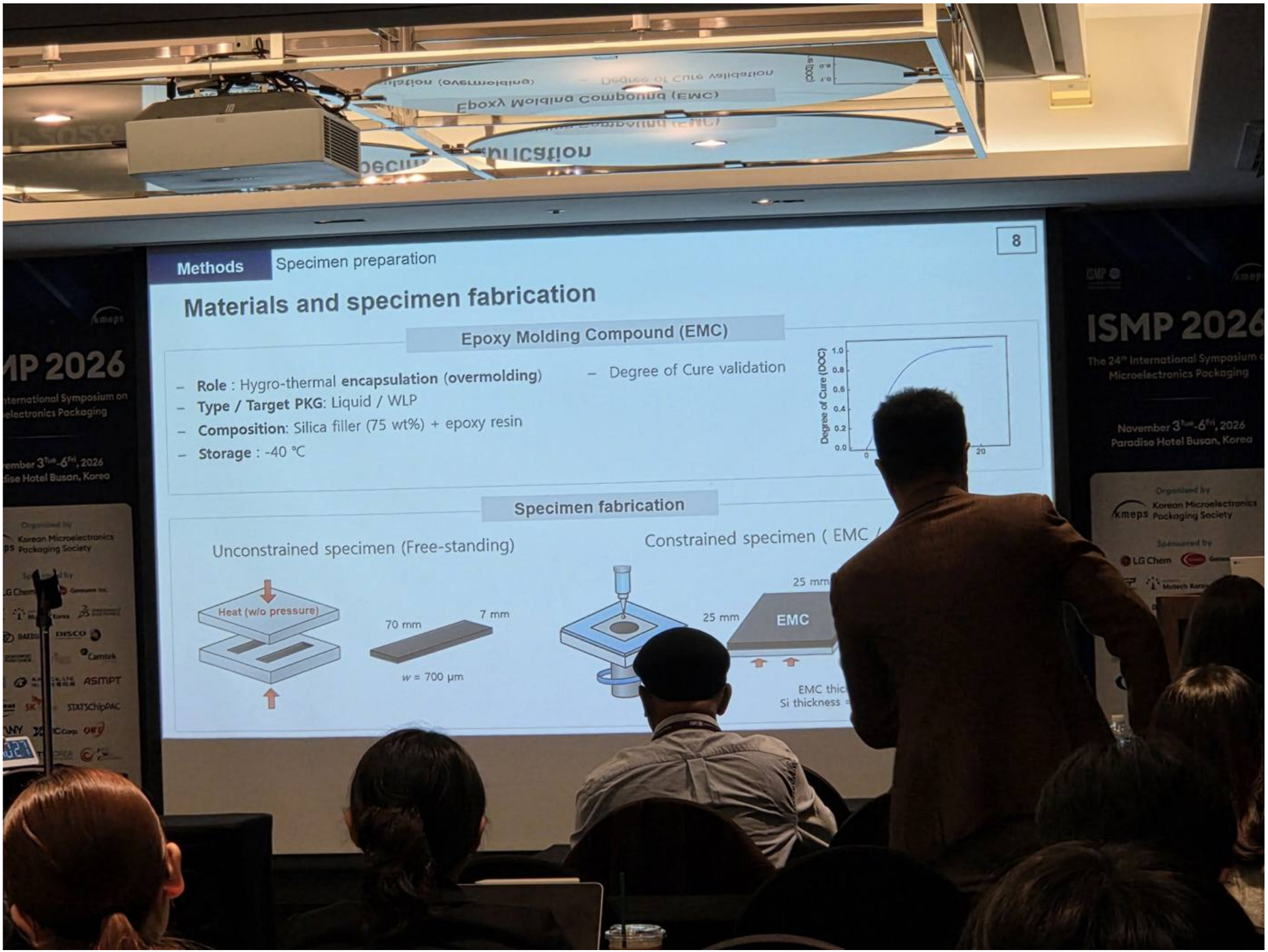
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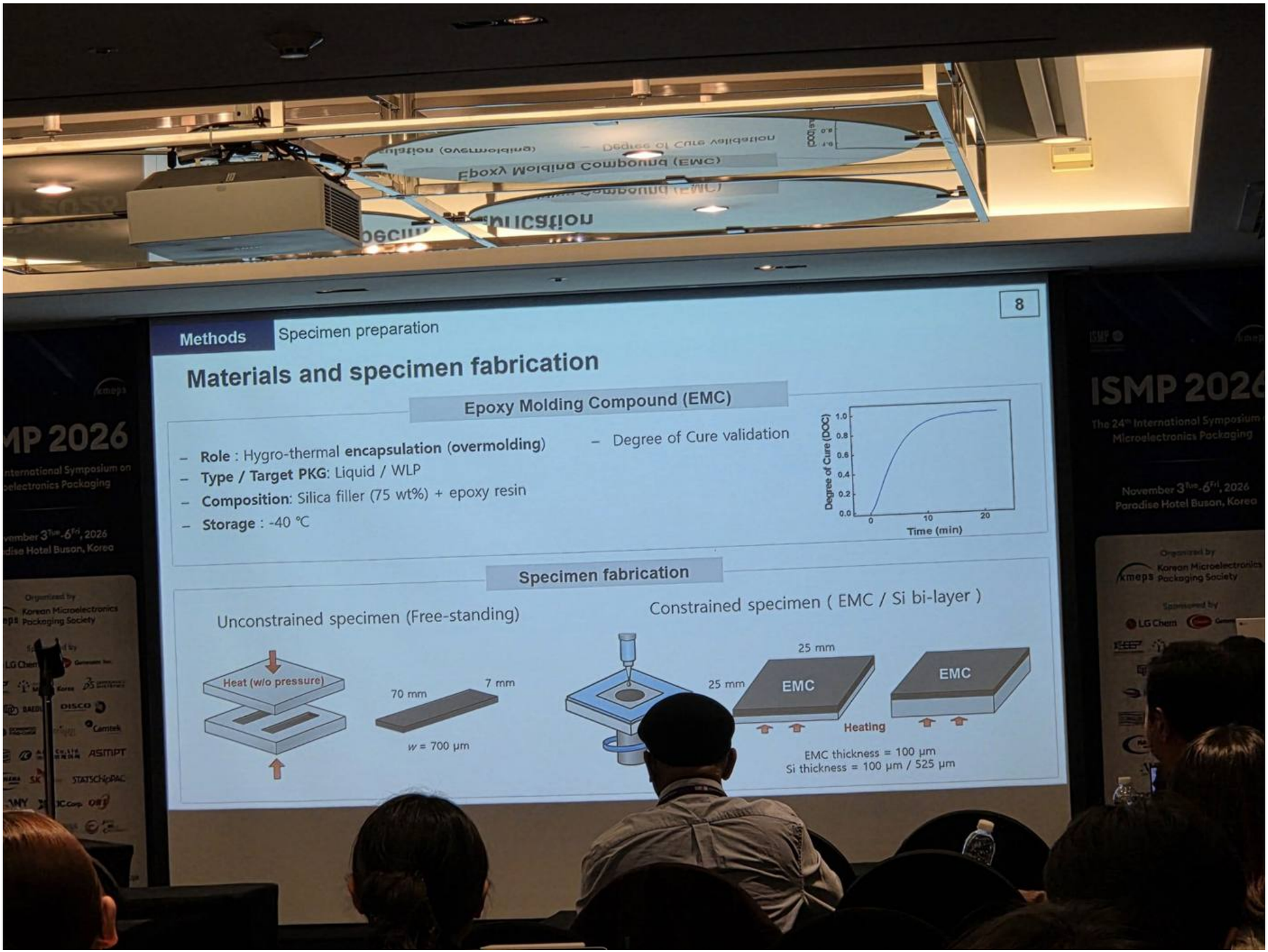
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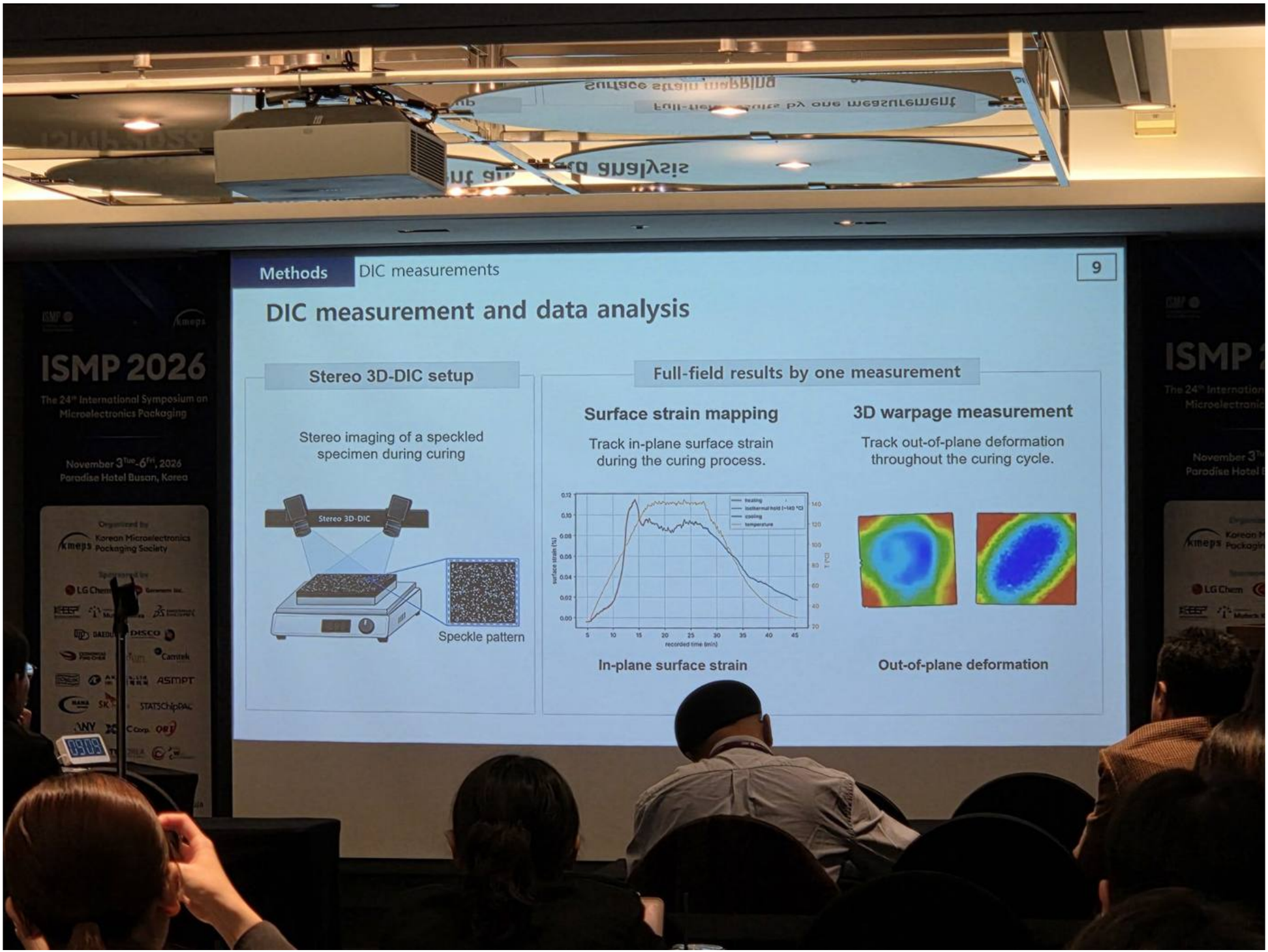
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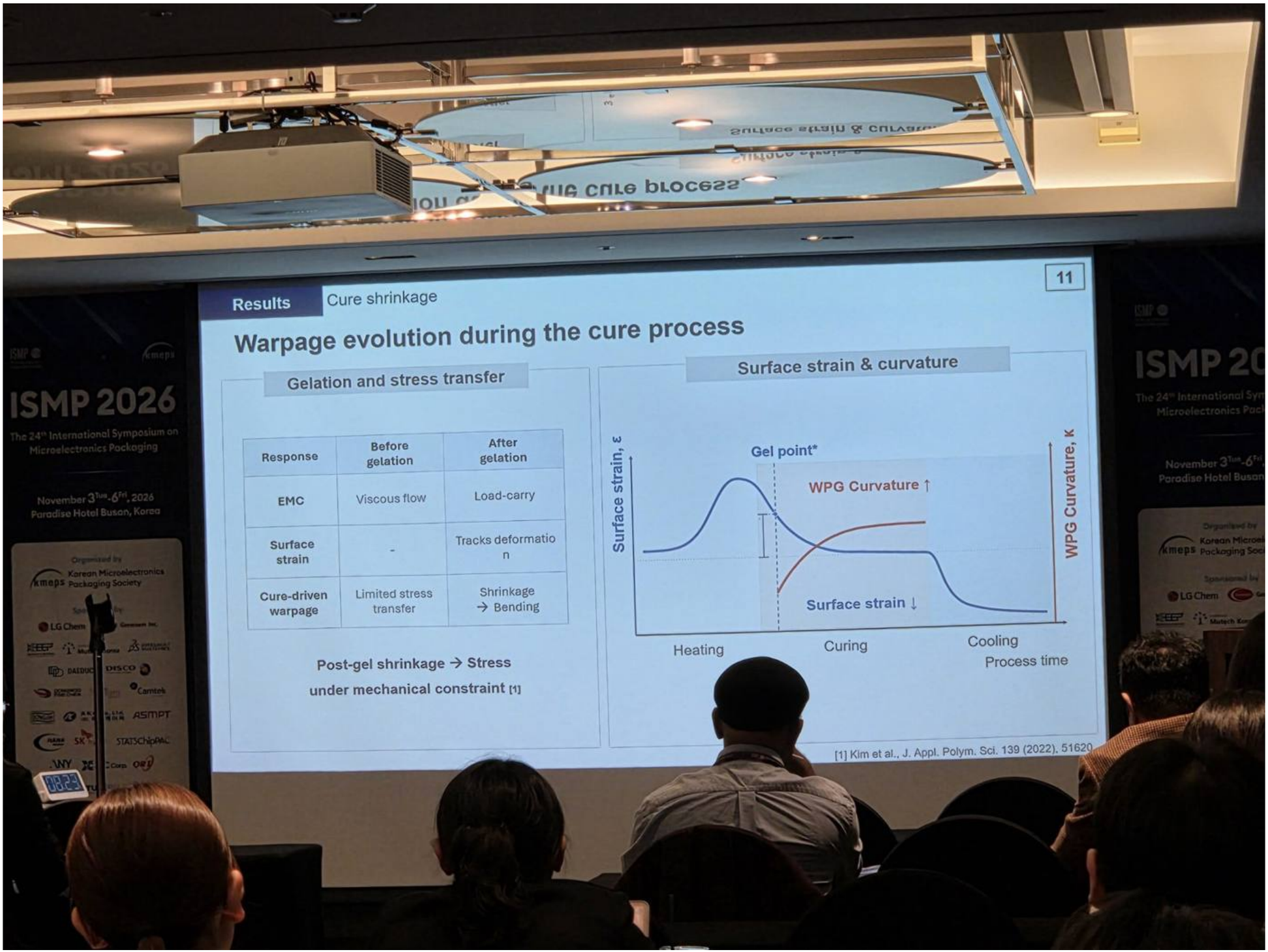
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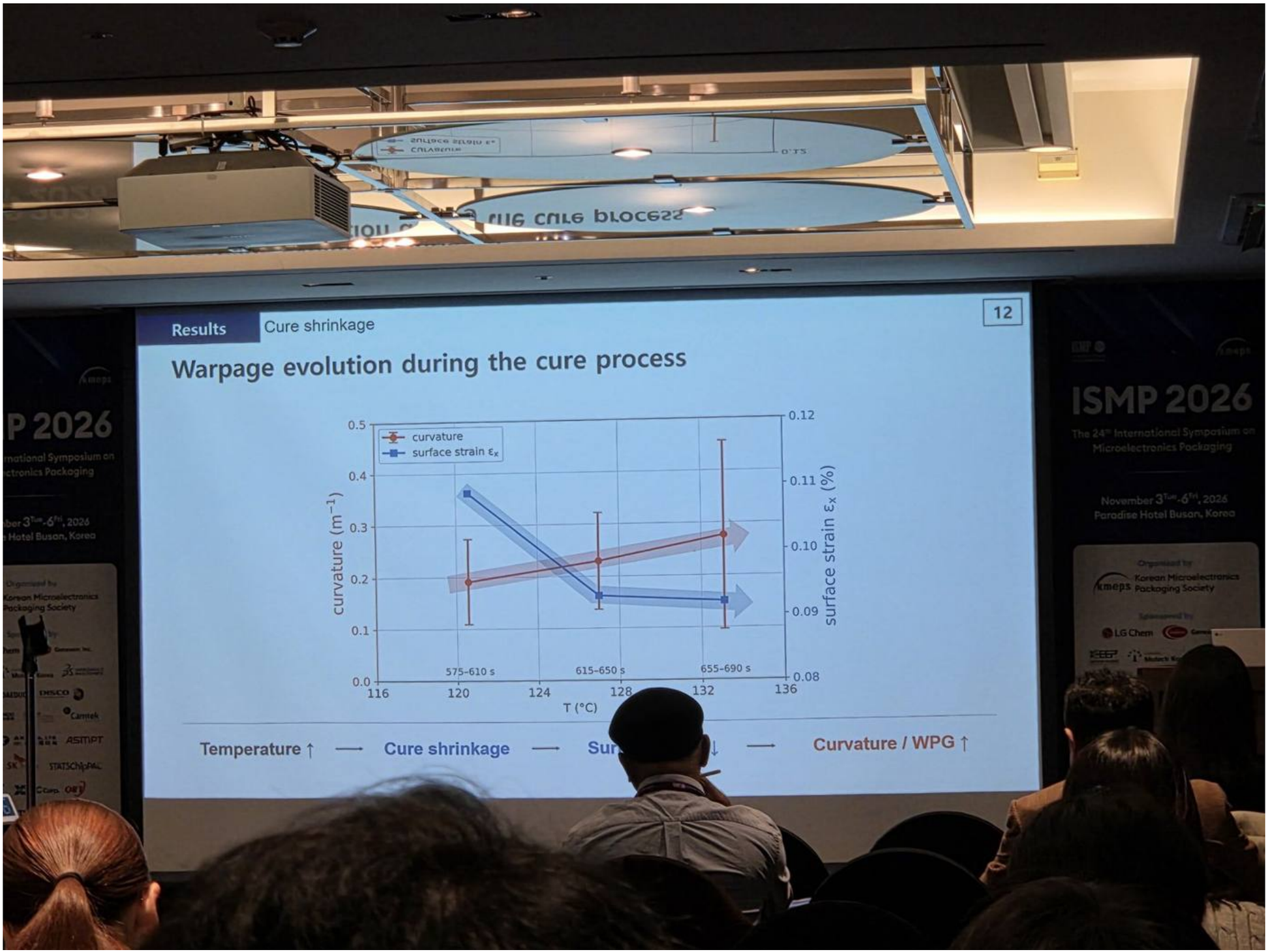
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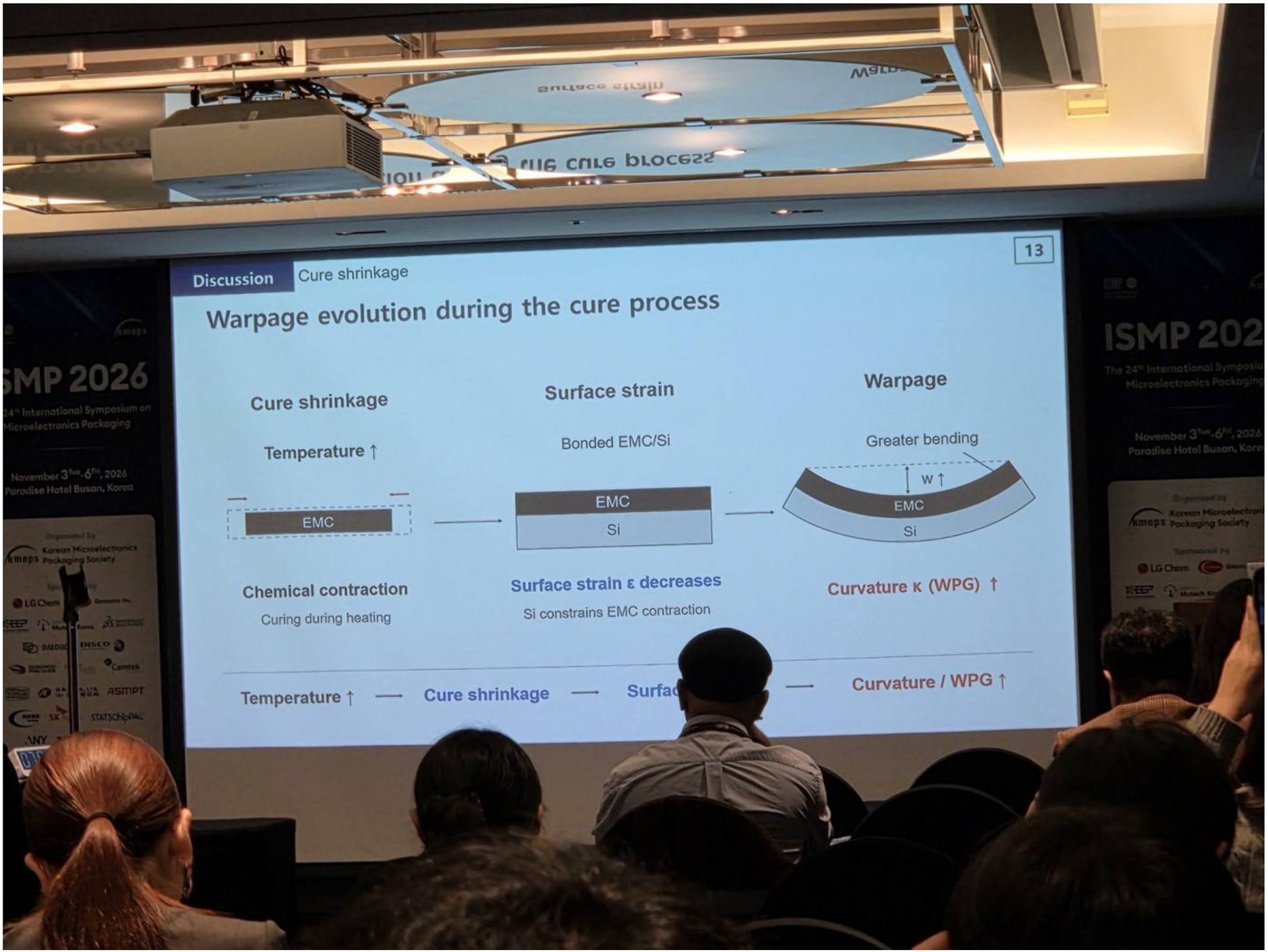
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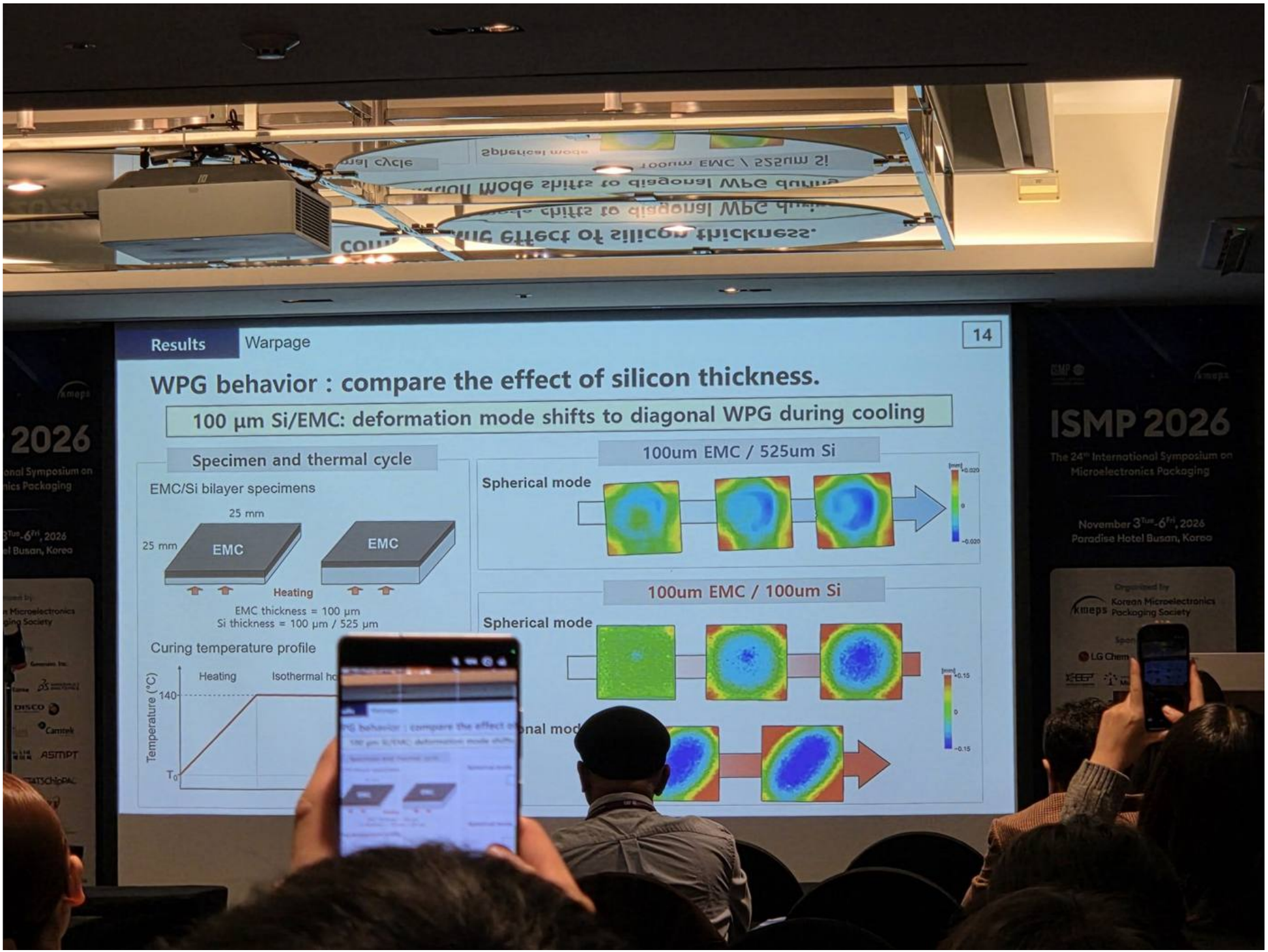
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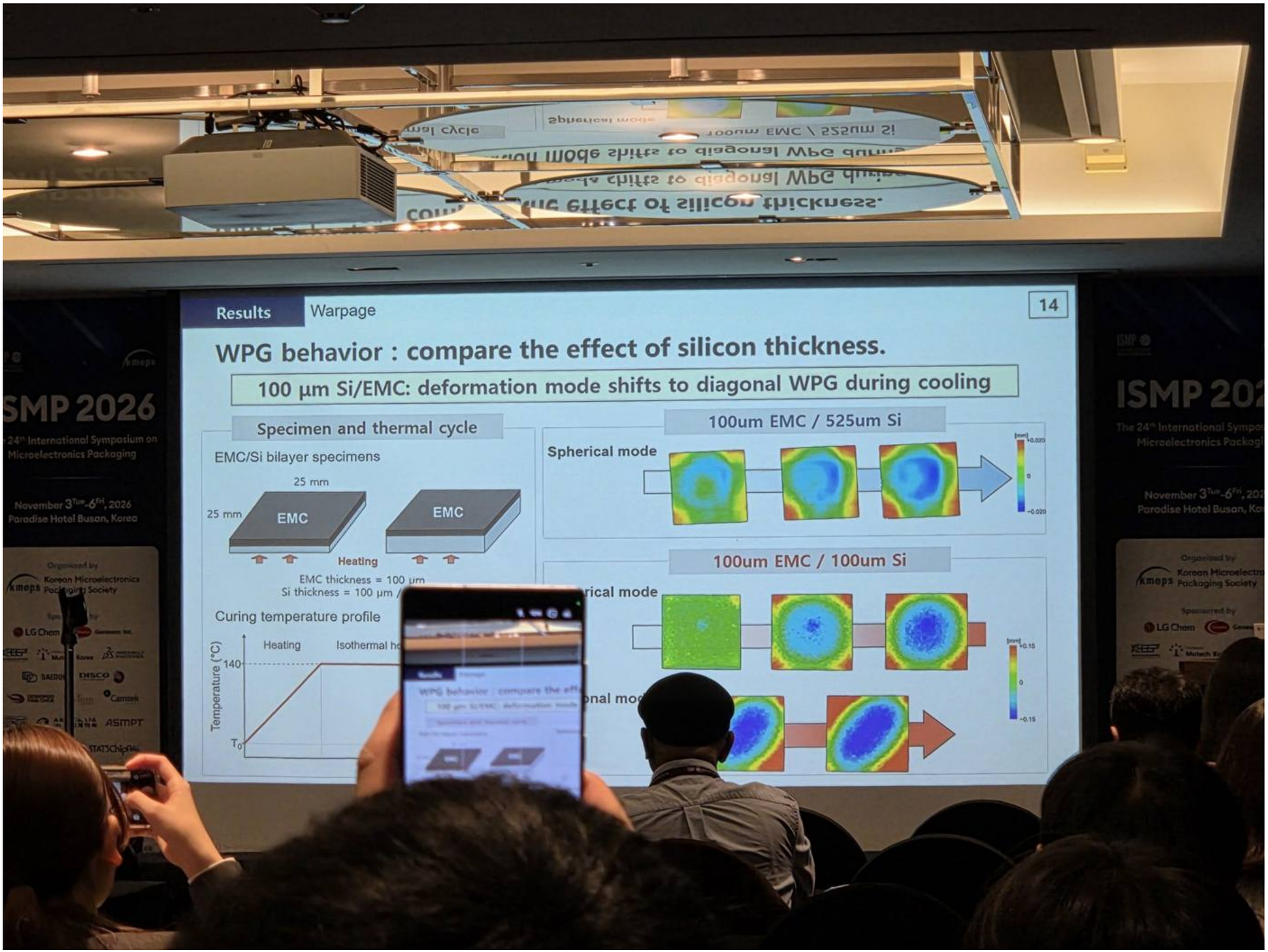
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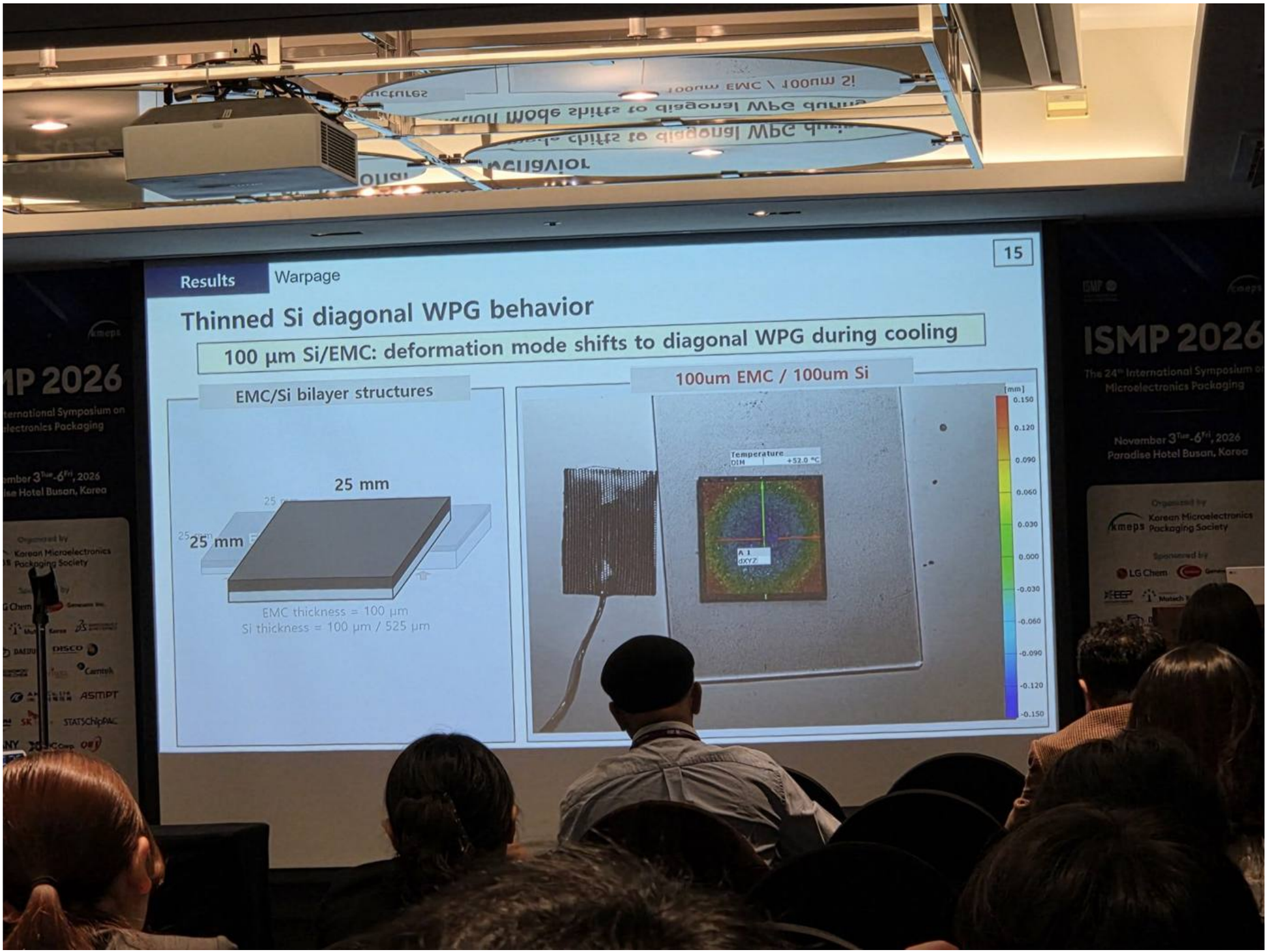
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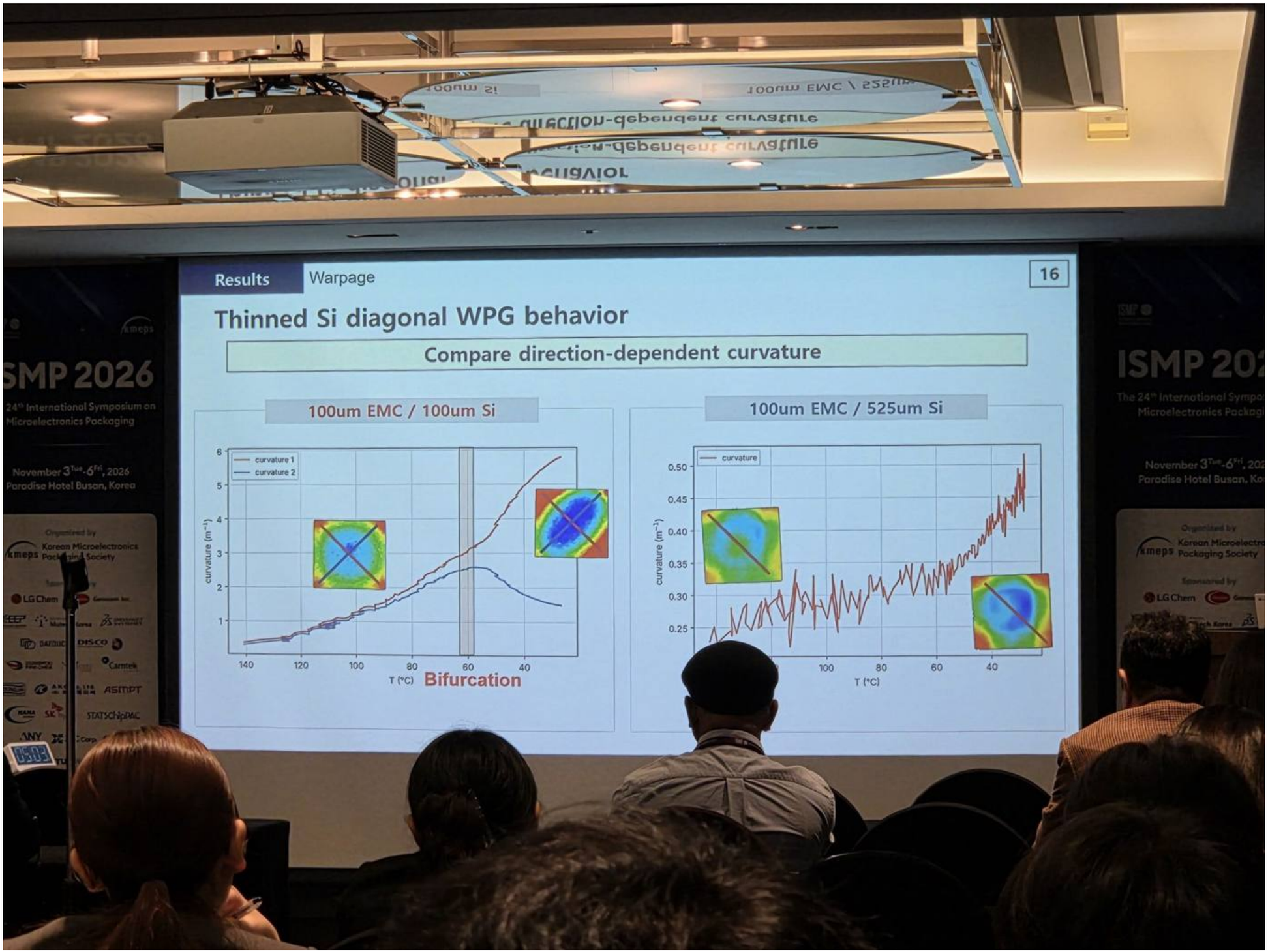
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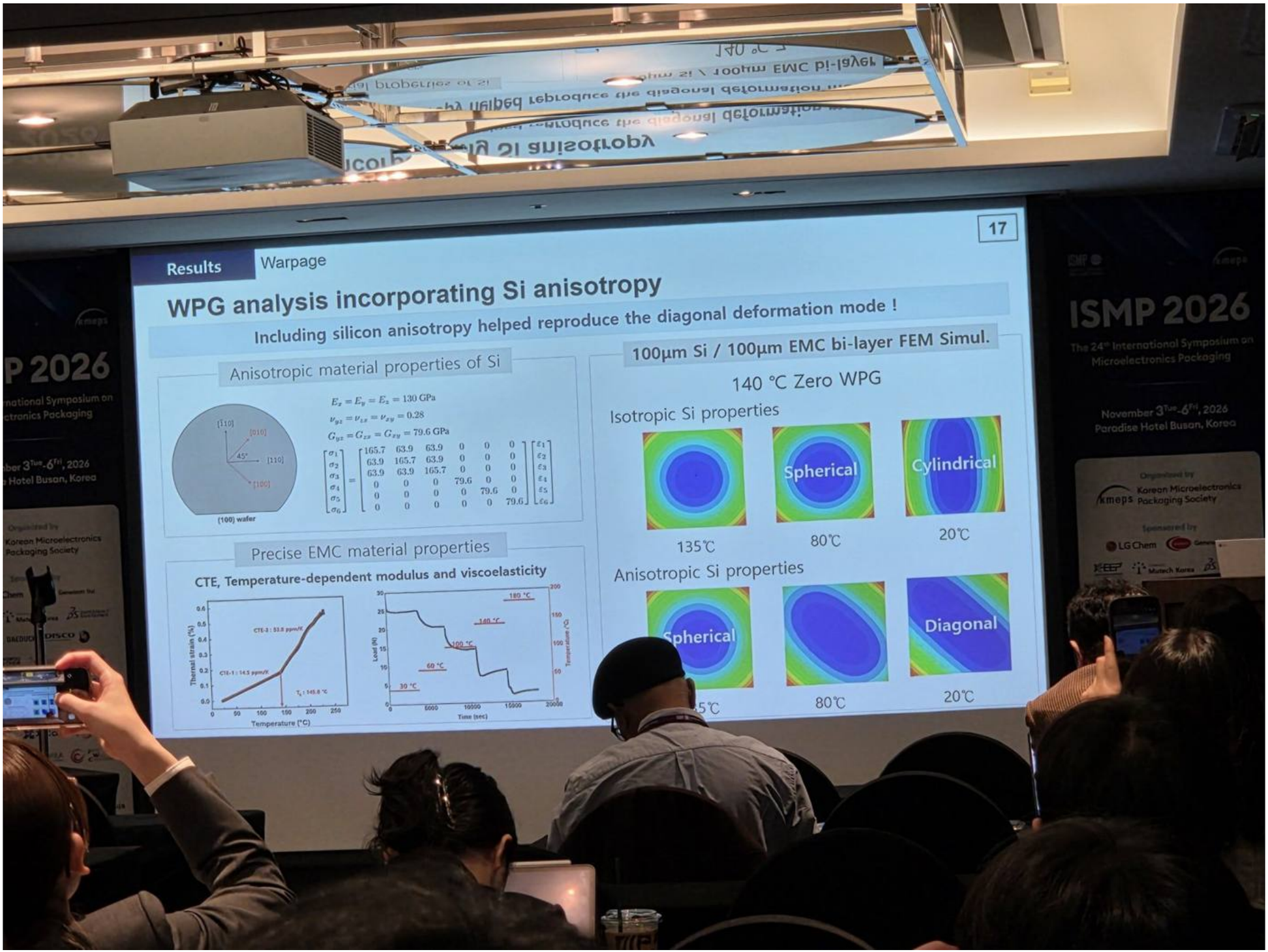
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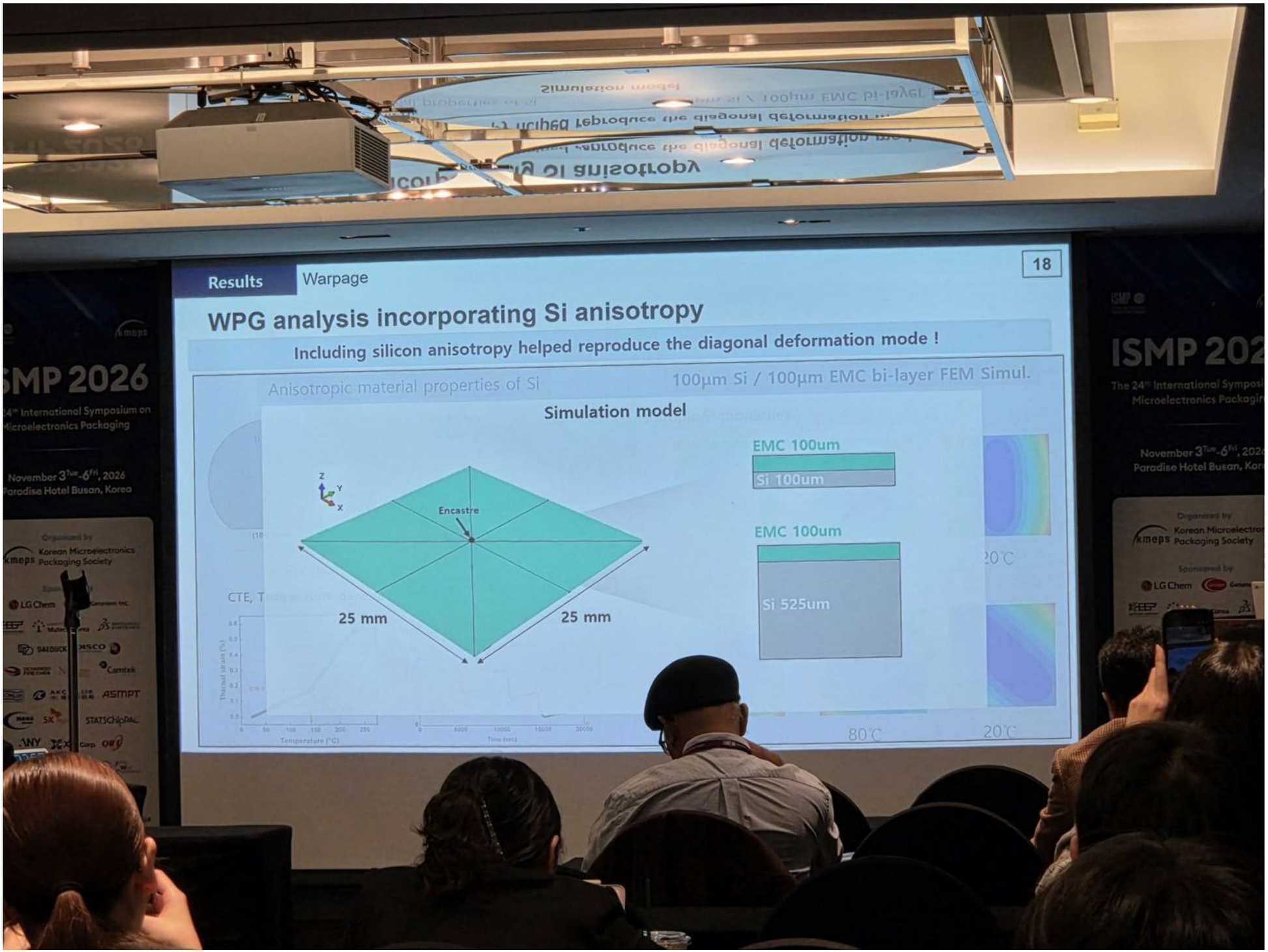
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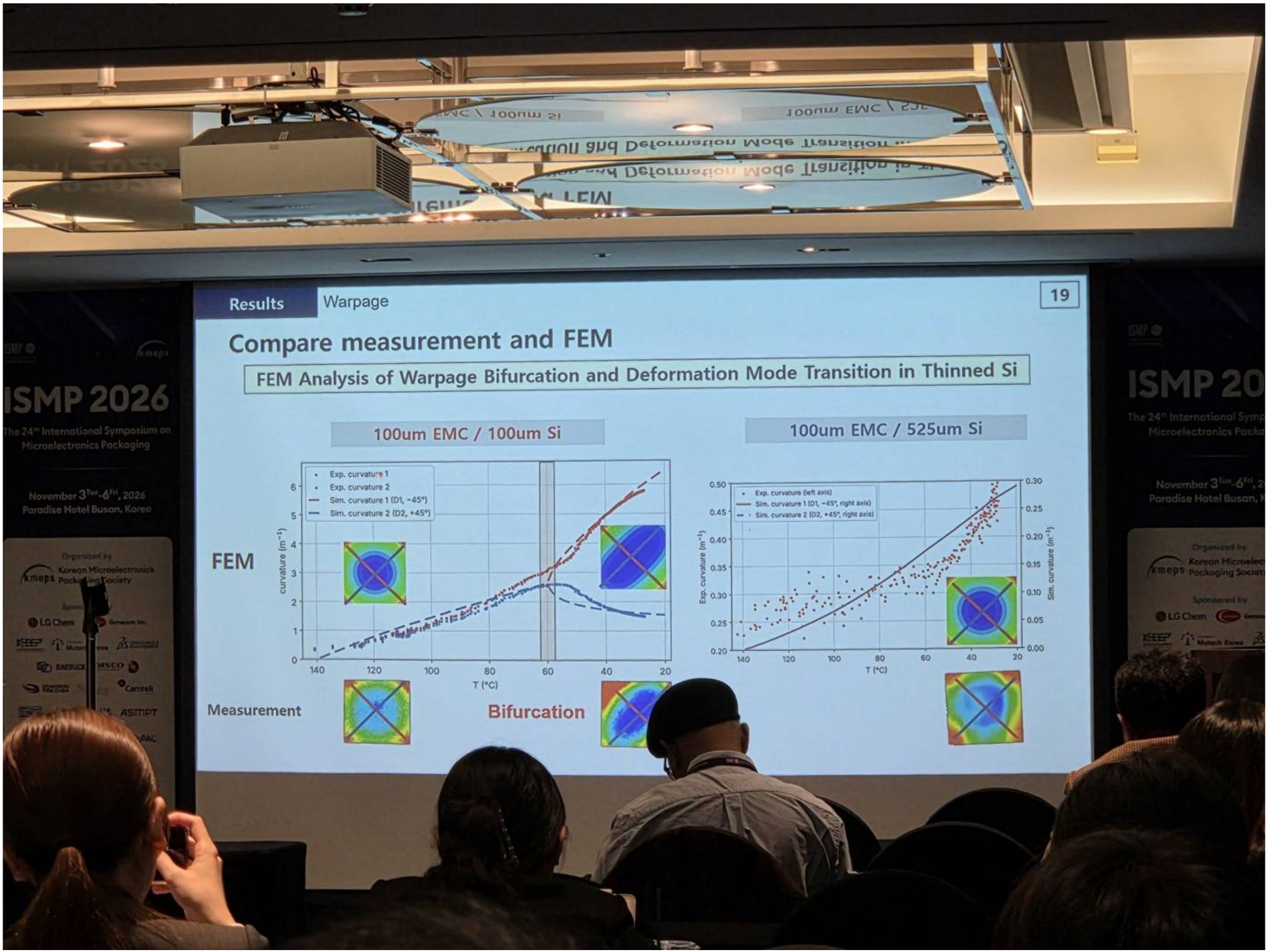
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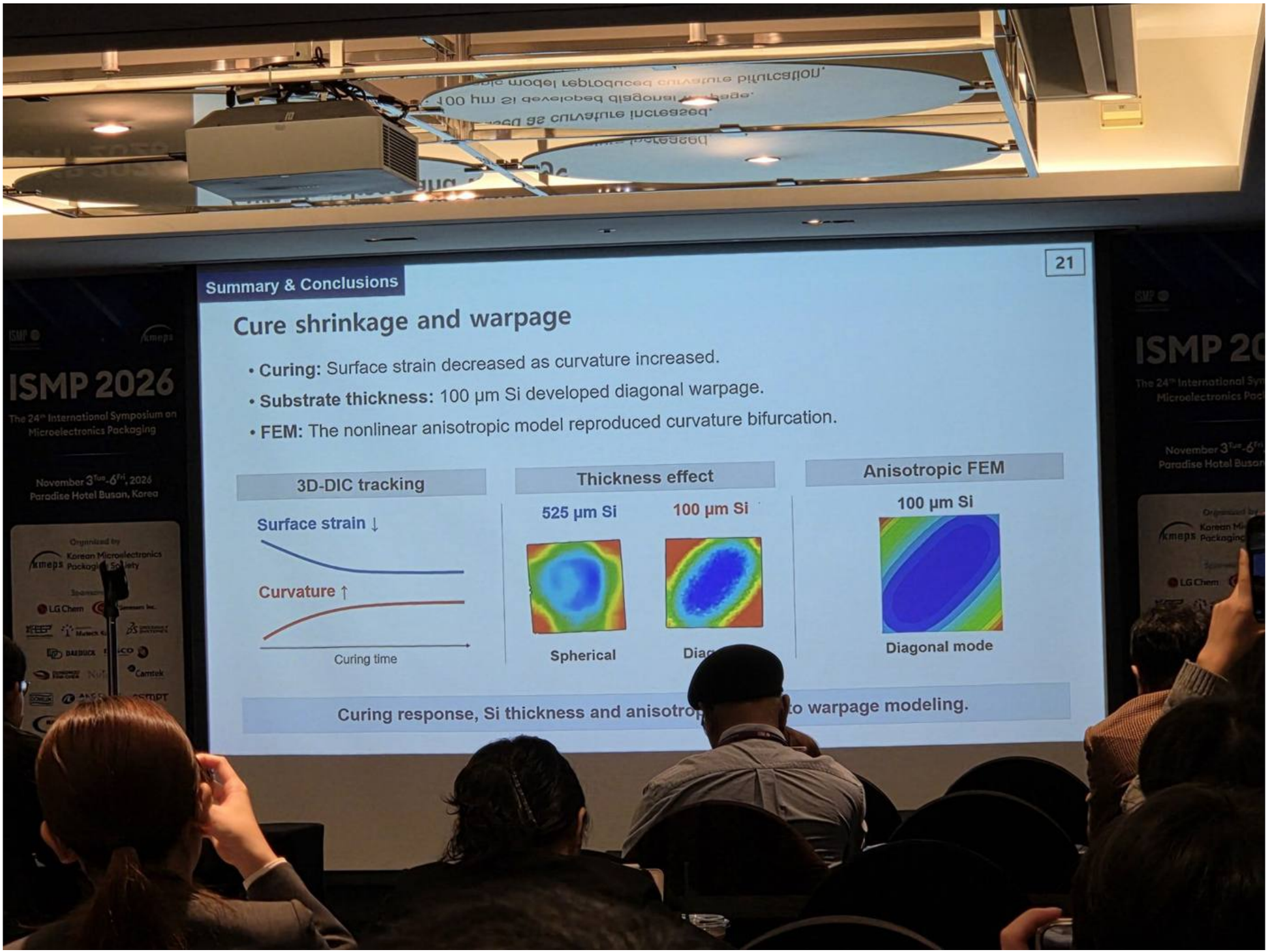
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T13 EMC 경화수축과 얇은 실리콘의 휨 분기

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Introduction

6

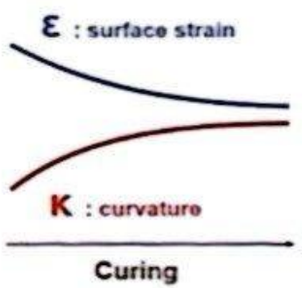
Research objective

Simultaneous strain–warpage tracking for improved EMC/Si warpage prediction

1. Track curing response by 3D-DIC

Surface strain + warpage





2. Compare warpage modes

Si/EMC bi-layer

(EMC thickness 100μm)



525 μm Si

100 μm Si

CamScanner로 스캔하기

3. Explain diagonal warpage by FEM



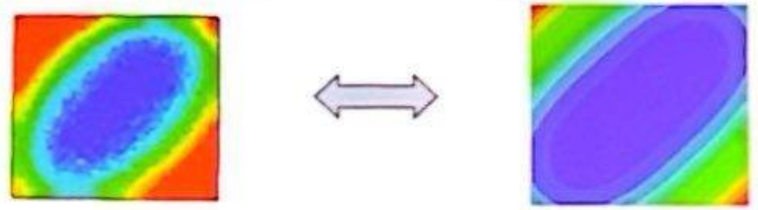
Si anisotropy

Large deformation

Compare curvature and deformation mode

Measurement

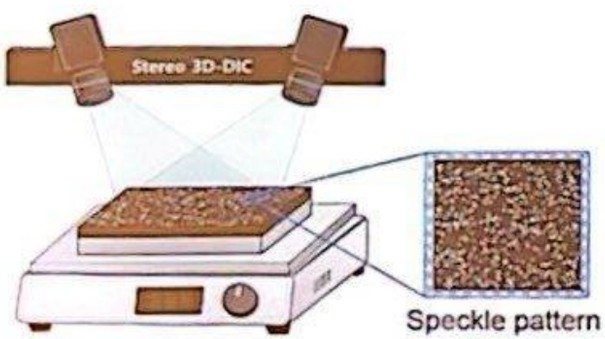
FEM



DIC measurement and data analysis

Stereo 3D-DIC setup

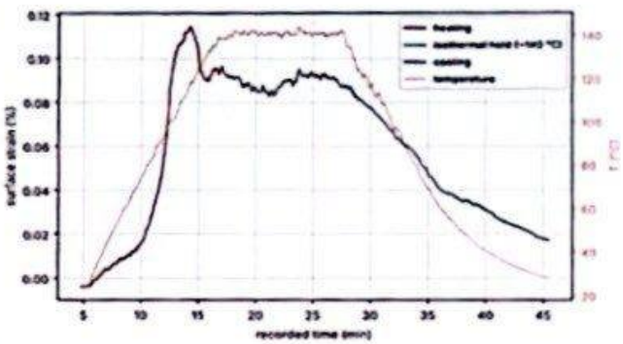
Stereo imaging of a speckled specimen during curing



Full-field results by one measurement

Surface strain mapping

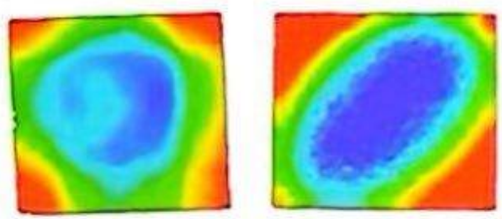
Track in-plane surface strain during the curing process.



In-plane surface strain

3D warpage measurement

Track out-of-plane deformation throughout the curing cycle.



Out-of-plane deformation

CamScanner로 스캔하기